

iPython (Google Colab)

[SWE2021]

JinYeong Bak

`jy.bak@skku.edu`

Human Language Intelligence Lab, SKKU

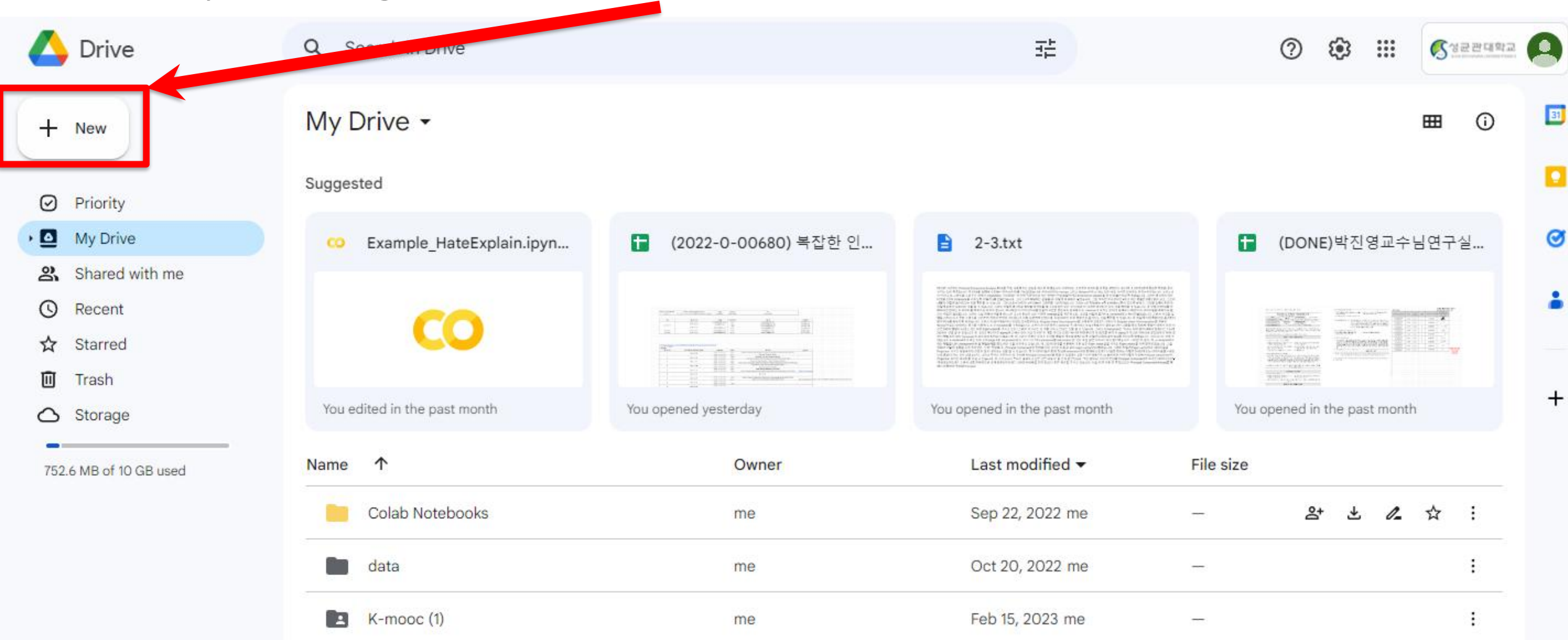
Slide credit: TaekHyun Kim (`treecko.kth@g.skku.edu`)

Introduction to Colab

- What is Google Colab?
 - Jupyter notebook development environment based on Cloud server
- Why use Colab?
 - Supports GPU for free (12hour session)
 - Accessible from everywhere because it is based on Cloud server
 - Easy to set up packages

How to Set up Colab

- On your Google Drive, click “+ New” button

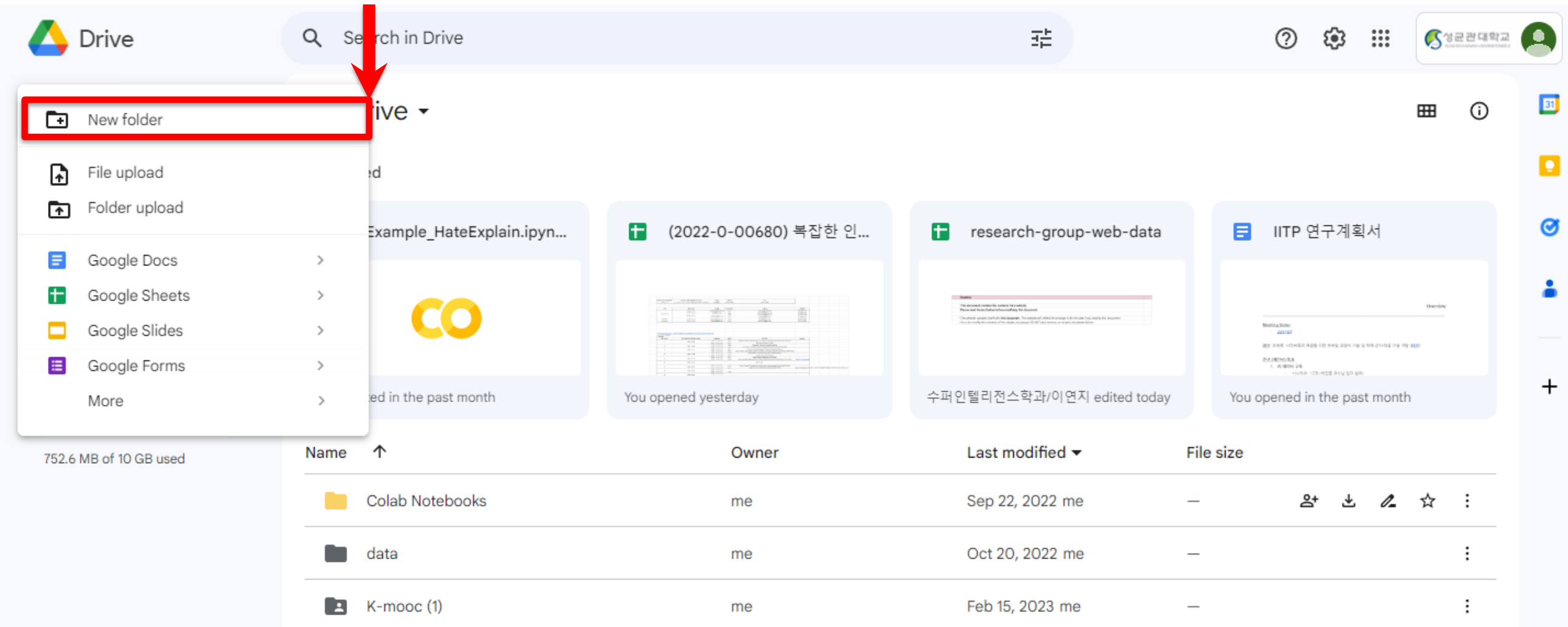


The screenshot shows the Google Drive web interface. A red arrow points to the “+ New” button in the left sidebar. The main area displays “My Drive” with a “Suggested” section containing four items: “Example_HateExplain.ipyn...”, “(2022-0-00680) 복잡한 인...”, “2-3.txt”, and “(DONE)박진영교수님연구실...”. Below this is a table listing files in the “Colab Notebooks” folder.

Name	Owner	Last modified	File size
Colab Notebooks	me	Sep 22, 2022 me	—
data	me	Oct 20, 2022 me	—
K-mooc (1)	me	Feb 15, 2023 me	—

How to Set up Colab

- Click “New folder” to make a project folder

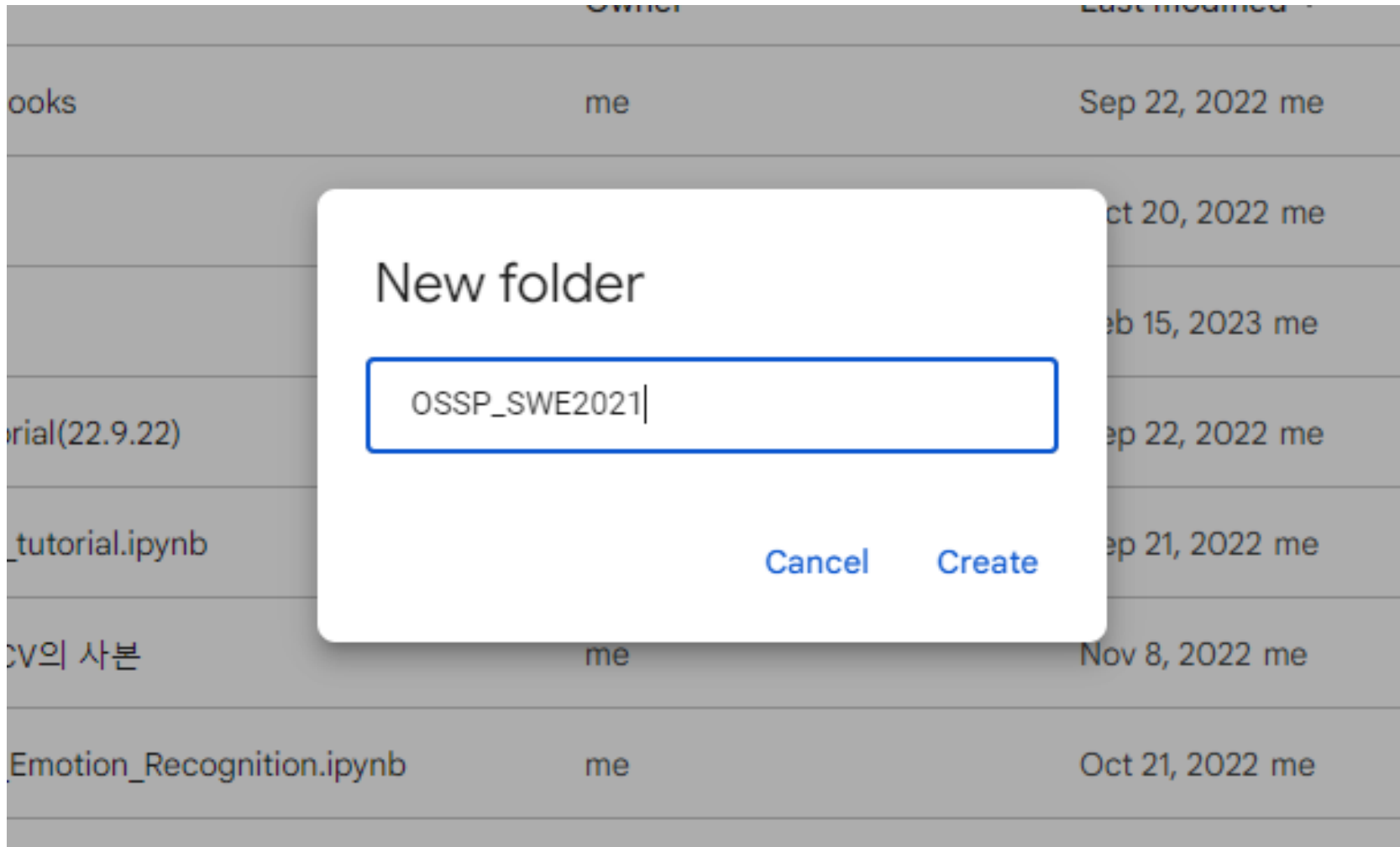


The screenshot shows the Google Drive web interface. At the top, there's a search bar and navigation icons. A dropdown menu is open, showing options like 'New folder', 'File upload', 'Folder upload', and Google Docs/Sheets/Slides/Forms. The 'New folder' option is highlighted with a red box, and a red arrow points to it from the text above. Below the menu, there are thumbnails of recent files and a table of files.

Name	Owner	Last modified	File size
Colab Notebooks	me	Sep 22, 2022	me
data	me	Oct 20, 2022	me
K-mooc (1)	me	Feb 15, 2023	me

How to Set up Colab

- Make a new folder with the name “OSSP_SWE2021”



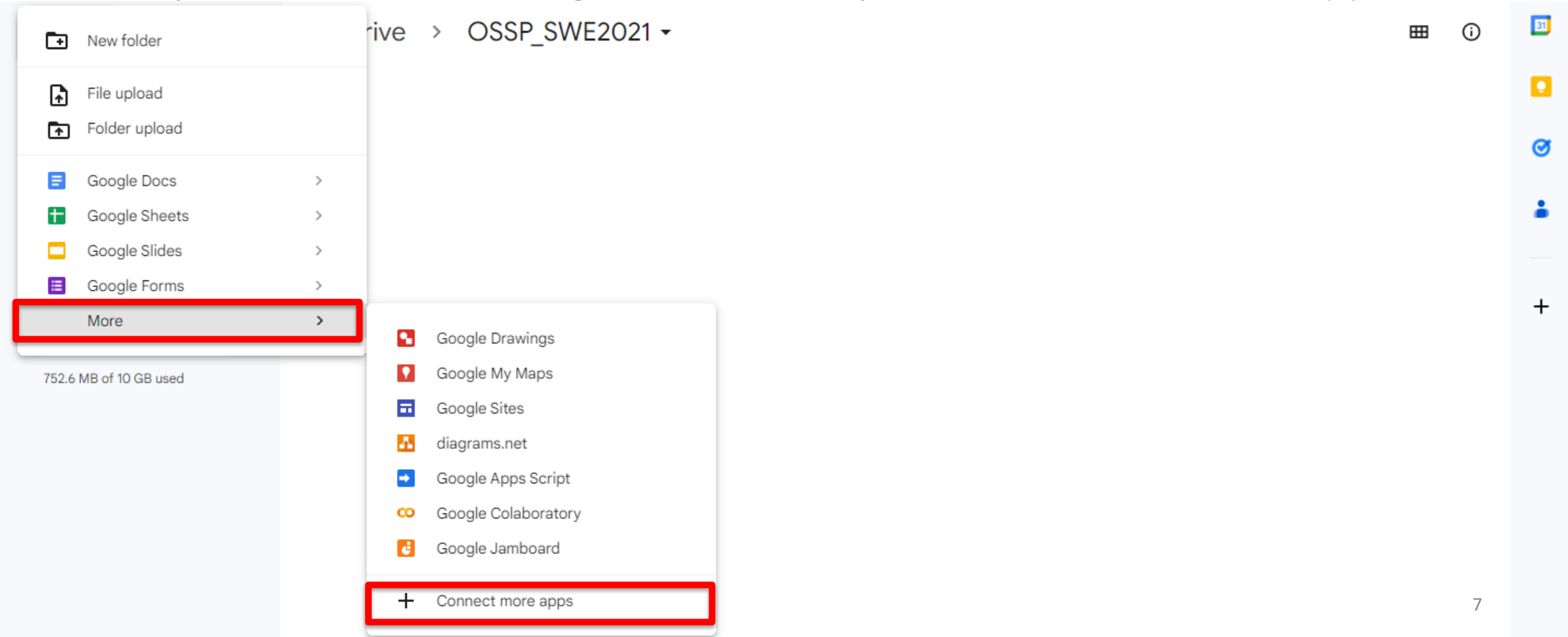
How to Set up Colab

- Click “Google Colaboratory” to make a new one



How to Set up Colab

- If you cannot find “Google Colaboratory”, Click **“Connect more apps”**



How to Set up Colab

- Search “Colaboratory”, and install it

The screenshot shows the Google Workspace Marketplace interface. At the top, there's a search bar with 'Colaboratory' entered. Below the search bar, there are filter buttons: 'All Filters', 'Works with', 'Price', and 'Internal apps'. The search results are titled 'Search results for Colaboratory' with a note that Google doesn't verify reviews or ratings. The first result, 'Colaboratory' by the Colaboratory team, is highlighted with a red box and has an 'Installed' badge. The other three results are 'MAIL MERGE for SHEETS' by yamm, 'Form Publisher' by Talarian, and 'Awesome Table - Data ...' by Talarian.

Google Workspace Marketplace

Colaboratory

All Filters Works with Price Internal apps

Search results for Colaboratory

Google doesn't verify reviews or ratings. [Learn more about reviews and results](#)

Colaboratory
Colaboratory team
This allows Google Colaboratory to open and create files in Google Drive. It is automaticall...
★ 4.7 • 10,000,000+

MAIL MERGE for SHEETS
yamm
Yet Another Mail Merge:...
Talarian
Send mass personalized emails with your Gmail™ account. Track engagement of each email in...
★ 4.7 • 10,000,000+

Form Publisher
Generate Powerpoints
Talarian
Generate templated documents with Google Forms. Generate PDF, Google Docs™, Sheets™, ...
★ 4.6 • 10,000,000+

GET DATA INTO SHEETS
Awesome Table - Data ...
Talarian
Export data from many sources to Google Sheets™. No technical skill required.
★ 4.5 • 10,000,000+

How to Set up Colab

The image shows two overlapping screenshots. The top screenshot is the Google Colab interface, displaying a notebook titled 'Untitled0.ipynb' with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a status bar indicating 'Last saved at 5:00 PM'. The bottom screenshot is the Google Drive interface, showing a search bar and a file list. The file list is filtered to show 'My Drive > OSSP_SWE2021'. A table of files is displayed, with the file 'Untitled0.ipynb' highlighted by a red rectangle. The table columns are Name, Owner, Last modified, and File size.

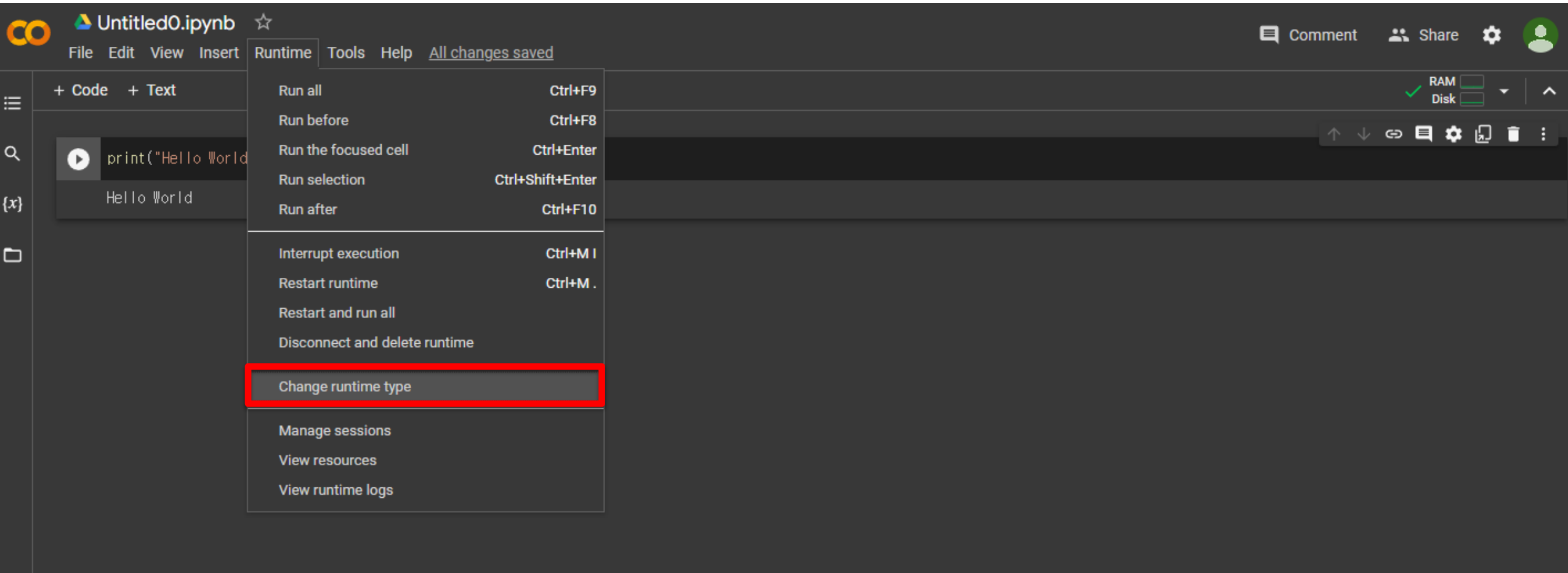
Name	Owner	Last modified	File size
Untitled0.ipynb	me	4:56 PM me	324 bytes

Let's Run Code

The screenshot displays a Jupyter Notebook interface with a dark theme. At the top, the title bar shows 'Untitled0.ipynb' with a star icon. Below it is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help', followed by the status 'All changes saved'. On the right side of the title bar are icons for 'Comment', 'Share', settings, and a user profile. The main workspace contains a single code cell. The cell's header has a play button icon, a green checkmark, and '0s'. The code inside the cell is `print("Hello World")`. Below the code, the output 'Hello World' is displayed. To the right of the code cell, there are icons for scrolling, linking, commenting, settings, copying, deleting, and a menu. In the top right corner of the workspace, there are RAM and Disk usage indicators, both showing green bars and a dropdown arrow. On the left side of the interface, there is a sidebar with icons for a menu, search, variables (showing '{x}'), and a file explorer.

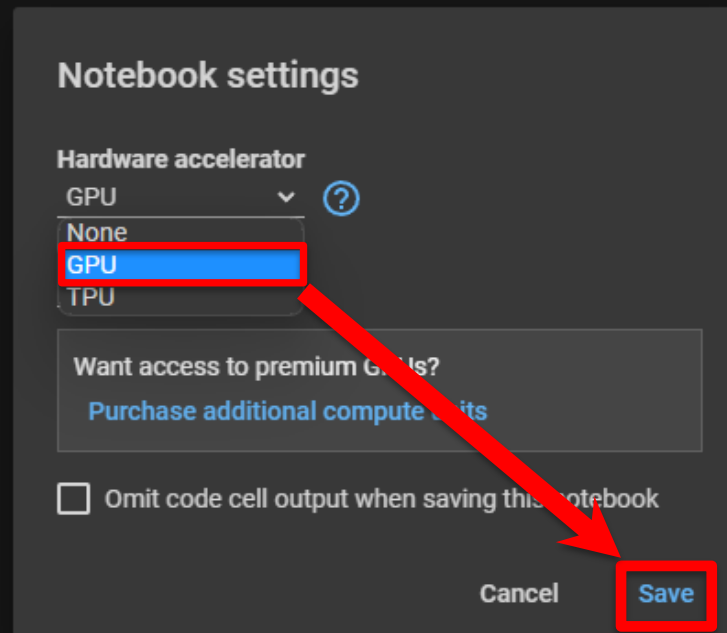
How to Change Runtime Type

- Change runtime type first in order to use GPU



How to Change Runtime Type

- Change Hardware accelerator into “GPU”, and click “Save”



Python Programming Basic – Data Type

- Go to a Google Colab page

<https://colab.research.google.com/drive/1-Hwru0Pdg-F4NFECuKesc71kES2a48LZ?usp=sharing>

- Copy the notebook to your Google drive
 - File => Save a copy in Drive
- Run the codes
 - Runtime => Run all
 - Click the play button on each code area
- Change the code and see the results

Python Programming Basic – If Else

- Go to a Google Colab page

<https://colab.research.google.com/drive/1P6HPHRPnnobTiKvjBTJqwzBdlBKAmNz2?usp=sharing>

- Copy the notebook to your Google drive
 - File => Save a copy in Drive
- Run the codes
 - Runtime => Run all
 - Click the play button on each code area
- Change the code and see the results

Python Programming Basic – Loops

- Go to a Google Colab page

https://colab.research.google.com/drive/1JMTDKoxPMzULKWr9xR8GvK_CvmvBLJuv?usp=sharing

- Copy the notebook to your Google drive
 - File => Save a copy in Drive
- Run the codes
 - Runtime => Run all
 - Click the play button on each code area
- Change the code and see the results

Python Programming Basic – Functions

- Go to a Google Colab page

<https://colab.research.google.com/drive/1TV2oEbCWQuwF6dnys1MBh7Zsr2l85vi3?usp=sharing>

- Copy the notebook to your Google drive
 - File => Save a copy in Drive
- Run the codes
 - Runtime => Run all
 - Click the play button on each code area
- Change the code and see the results

Python Programming Basic – File Handling

- Go to a Google Colab page

https://colab.research.google.com/drive/1a_d5iET_3p_2ToYwloT5snFasLsGEopU?usp=sharing

- Copy the notebook to your Google drive
 - File => Save a copy in Drive
- Run the codes
 - Runtime => Run all
 - Click the play button on each code area
- Change the code and see the results

Reading Materials

- W3schools Python Tutorial (English)
 - <https://www.w3schools.com/python/default.asp>
- Jump to Python (Korean)
 - <https://wikidocs.net/book/1>
- (Recommendation) To study more, check “Try/Except”, “Classes/Objects” parts by yourself